Loïc Strullu 2023

Practical Work 1: The JavaSTICS interface

We are going to create a simulation with the STICS model. We want to simulate a winter wheat following a spring pea crop at Mons in the Picardie region in Northern France. The simulation begins in summer the 22nd July 2010 after the harvest of spring pea in 21st July 2010. The spring pea produced 15.2 t ha^{-1} of aerial biomass and 7.4 t ha^{-1} of grain.

I. Create a climate file and the associated weather station file

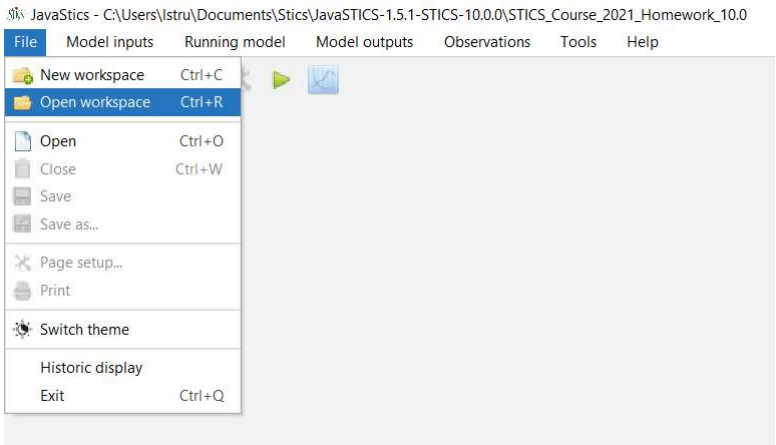
We have climatic data for the years 2010 and 2011 measured near the experimental site by a weather station of INRAE. Climatic data are from the INRAE database Climatik®. Mandatory variables are minimum temperature, maximum temperature, global radiation and rainfall. Optional variables are PET, wind and vapor pressure.

1. Open the file clim_mons_2010_2011.xlsx which is stored in the working directory Practical_Work_1

2. Replace colon by points
Edition → Replace → , by . → Replace all or use (ctrl + f)

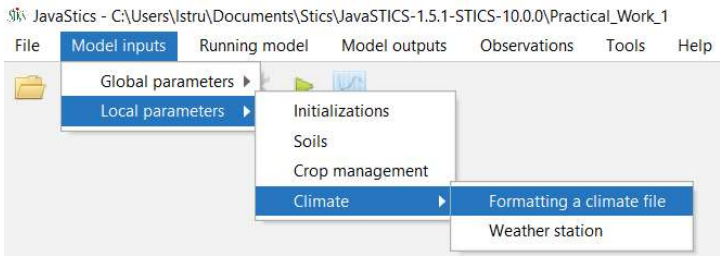
3. Save the file in csv format:
File → Save as → CSV file (separator: semicolon)

4. In STICS, launch JavaStics.exe.
5. Open the file tab and click on "open workspace"



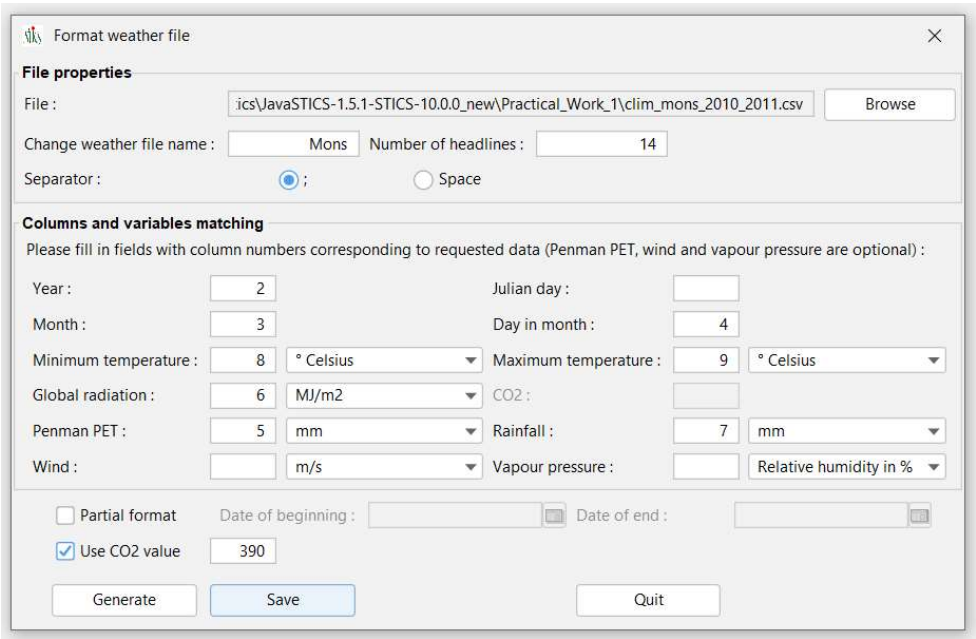
6. Select the working directory Practical_Work_1
7. How do I create a weather file?

/!\ Other ways possible, but the following one is convenient especially if we have a lot of climatic years.



- ✓ Give the pathway to the climate file
- ✓ Give a name to the weather file that will be created = Mons (/!\ has to be the same as the station name)
- ✓ Number of header rows = 14 → File information + name of variables
- ✓ Put the column number that corresponds for each input variable of the model, do not put anything if you do not have the variable.
- ✓ Select the correct units for each variable:

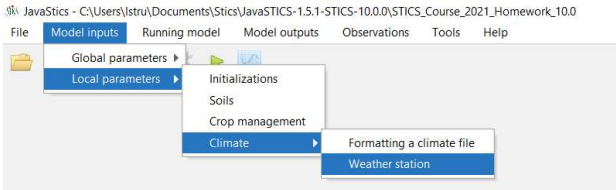
  Temperature (TN and TX) = °C
  Penman PET (ETPP) and precipitation (RR) = mm
  Radiation (RG) = megajoules m^{-2}
  Add a value for CO_2 = 390 ppm (possible to get CO_2 concentration values at: https://gml.noaa.gov/ccgg/trends/)



- ✓ Save
- ✓ Generate

8. Go to the working directory PW1. The weather files have been created = Mons.2010 and Mons.2011

9. How do I create a weather station file?

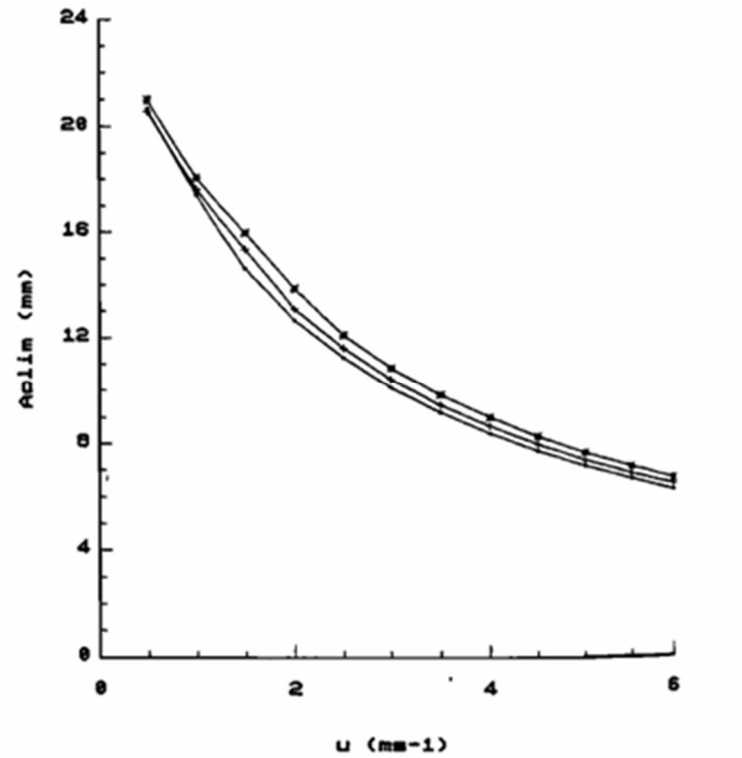


10. Click on "newfile" and give a name to the station file = Mons (/!\ same name as used for the climatic file)

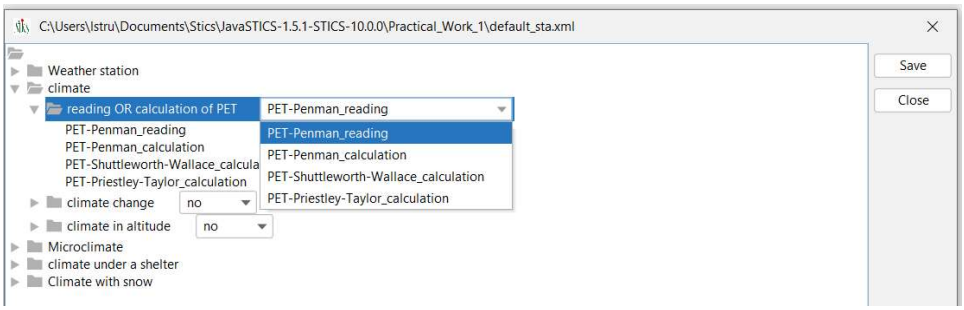
11. Open the file inputs_stics_PW1.xlsx at tab Station:
    - ✓ Open your station file and modify the latitude = 49°

- ✓ Note that mineral nitrogen content input through rainfall can be considered: parameter concRR (keep the default value 0.02 in our case)

- ✓ Set the value of the parameter aclim to 11.8 (function of average wind velocity the days without rain).



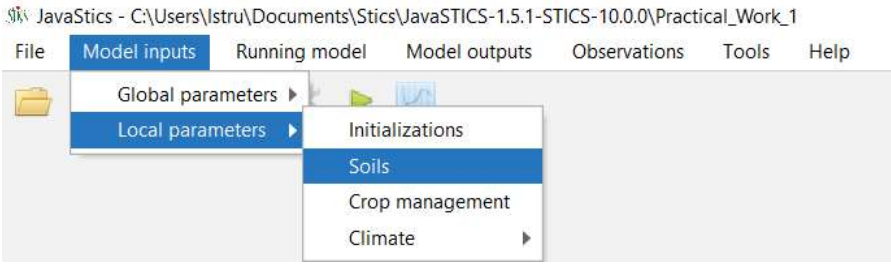
- ✓ Choice of PET calculation in the "climate" tab: depends on the weather data available!



- ✓ Penman reading: this is our case, we have this data as an input variable
- ✓ Penman calculate: in this case two additional weather variables are mandatory (wind speed and vapour pressure)
- ✓ Shutwall and Wallace: in this case two additional weather variables are mandatory (wind speed and vapour pressure). This method of PET calculation is mandatory for row crops and intercrop.
- ✓ Prestley-Taylor: if you don't have the ETP or wind speed and vapour pressure deficit

II. Create a soil file

1. In JavaSTICS:



2. Create a new soil (Add) → name = Mons.

3. Fill the parameters value thanks to the parameters value given in the tab Soils of the inputs_stics_PW1 excel file

4. In STICS you can have 5 soil layers at most, and this is important to think about what layers you will describe (depend on the soil description, available measurements, kind of processes you are most interested in, …)

Some help to estimate albedo, Q0, epd is available below:

For parameter Q0:

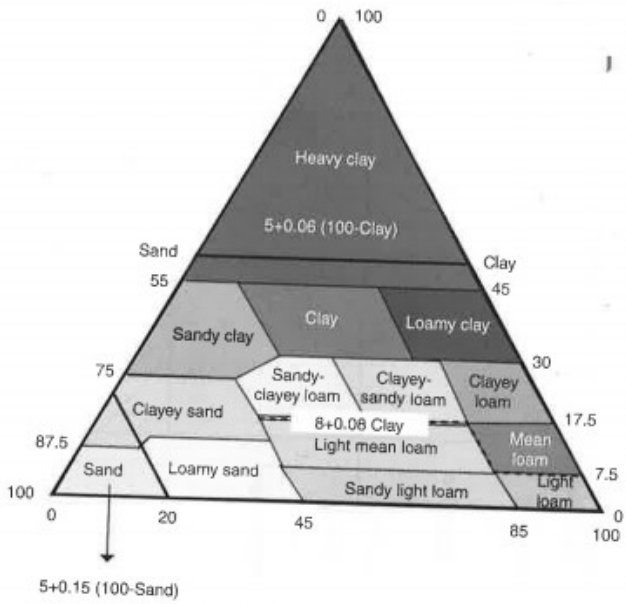


Figure 11.6. Textural triangle and classification by Jamagne *et al.* (1977). Soil transfer functions to assess the $Q0_s$ parameter as a function of clay or sand content.

Type de sol	q0	source
argi < 5	6	Ritchie, 1972
5< argi < 25	9	Ritchie, 1972
25< argi <45	12	Ritchie, 1972
argi > 45	6	Ritchie, 1972
argi = 27	22	Brisson et Perrier, 1991
argi = 51	0	Brisson et Perrier, 1991

For albedo:

tabl. 6 : Values of the dry soil albedo ($ALBEDO_S$) using either textural classes or colours, based on Richard and Cellier (1998), Jacquemoud and Baret (1992) and http://www.icrisat.org/gt-aes/oneds/DataNeeds.htm

Soil type	$ALBEDO_S$
TEXTURE	
Limestone	0.31
Loamy sand	0.25
Clayey loam	0.18 – 0.22
Mean loam	0.22 – 0.23
Crusted mean loam	0.28
Clay	0.28
COLOUR	
Brown soil	0.27
Red soil	0.29
Black soil	0.17
Grey soil	0.29

For epd parameter which controls nitrate dispersion can be estimated from soil textural class and the αpred value in the table below: epd = 2 αpred (be careful, should be in cm in STICS!)

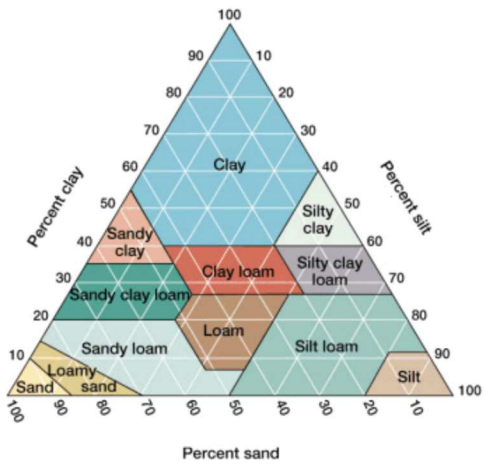


Table 6. Dispersivity (predicted using the regression coefficients in Table 5) as a function of soil textural class.†

Textural Class	n	ψ_a	b	α_{pred}
		kPa		10^{-3} m
Sand	14	0.69	2.79	8
Loamy sand	30	0.36	4.26	26
Sandy loam	124	1.41	4.74	34
Loam	103	3.55	5.25	46
Silt loam	394	7.59	5.33	56
Sandy clay loam	104	1.35	6.77	60
Clay loam	147	2.63	8.17	81
Silty clay loam	324	6.17	8.72	96
Sandy clay	16	0.98	10.73	109
Silty clay	43	3.24	10.39	110
Clay	148	4.68	11.55	128

† Mean values of ψ_a and b from Cosby et al. (1984).

5. SAVE !

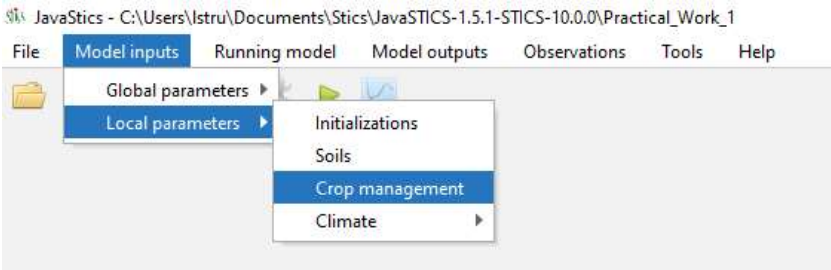
III. Create a technical file

1. We have retrieved the data of the technical itinerary applied to our culture, and here are presented the information we have:

Residues of previous crop: Straw of previous crop
Tillage:
06/08/2010 on 7 cm
16/09/2010 on 7 cm
21/10/2010 on 7 cm
Description of crop sowing:
Winter wheat variety Shango was sown the 21st October 2010 at 2.5 cm depth with a crop density of 200 plants m^{-2}
Fertilisation:
29/03/2011 110 kg ha^{-1} of UAN solution
04/05/2011 46 kg ha^{-1} of UAN solution
Harvest:
Crop was harvested the 1st August 2011 and only stubble were returned to the soil.

This information is summarized in the tab Tec of the inputs_stics_PW1 excel file

2. In JavaSTICS:



3. Create a new file with the name Wheat_2011 created from the default technical file

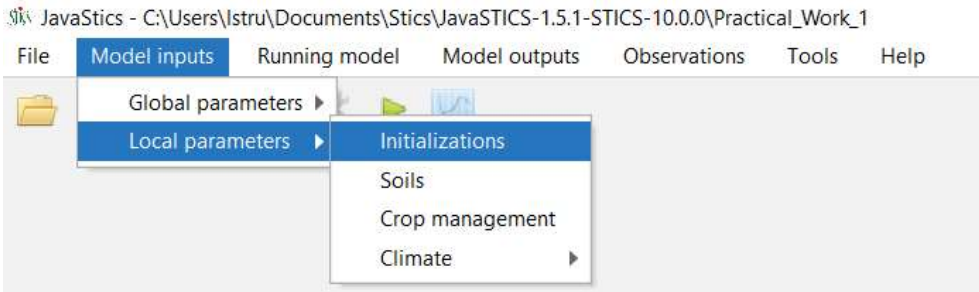
4. Fill the file with the information given earlier (/!\ you can use the tables given in the appendices of the JavaSTICS documentation to transform the given dates in julian days or use an excel formulae)

5. SAVE

IV. Create an initialization file

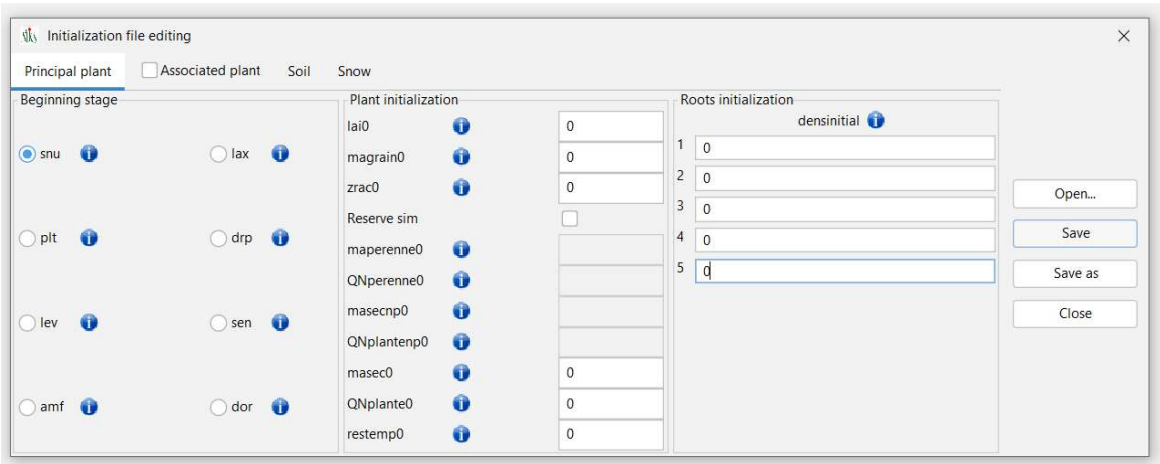
1. What can I do if I don't have measurements of water and nitrogen to initialize the simulation?
   - ✓ I have information about previous crops and management practices:
     I can launch chained simulations for a minimum of 2 years with a default initialization at the beginning and use model output data on water and nitrogen to initialize the model
   - ✓ I have information about the previous crop but not for management practices:
     Use a default initialization with average data of soil water and nitrogen content observed after cultivation of the previous crop
   - ✓ I have information about soil water and N content but not at the beginning of the simulation:
     You can do a calibration of your initial soil water and nitrogen content

2. In JavaSTICS:



3. Fill in the file thanks to information given in the tab Ini of the inputs_stics_PW1 excel file: Be careful to put dots to separate the decimals!

For the plant part, put 0 in the boxes, indeed the simulation begins in bare soil (snu).
/!\ Do not let an empty cell !

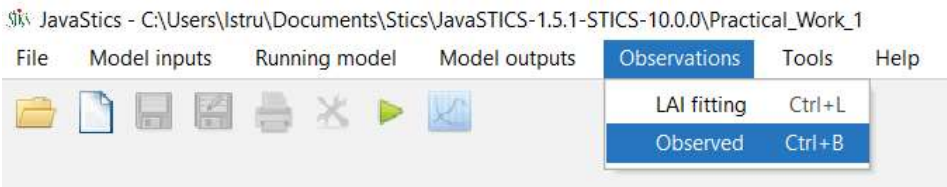


4. In the "Soil" tab, a value of 0 means 0 for NH4 and NO3 content, but is interpreted as field capacity for the soil water content /!\ Do not let an empty cell !
5. SAVE AS Wheat_2011_ini.xml

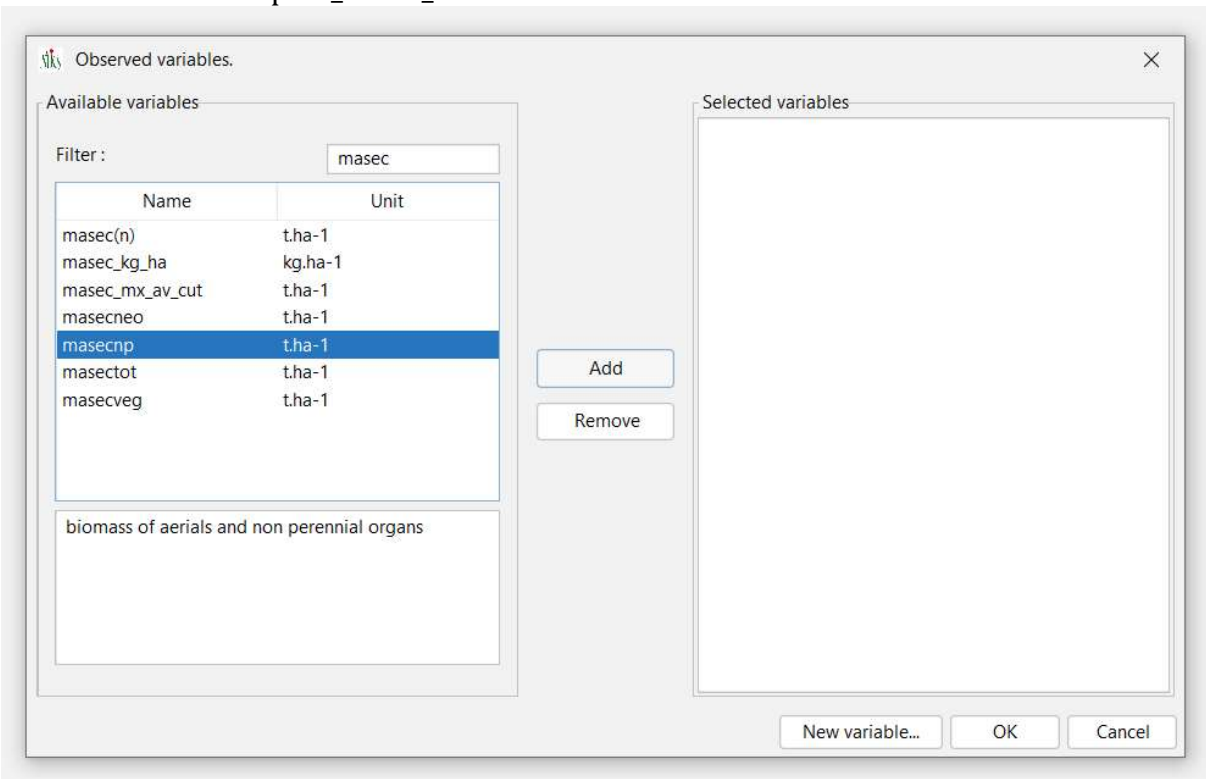
V. Create an observation file

We have data available for biomass production (aerial biomass and grain) at harvest and for water and nitrogen content at the end of winter. We are going to create an observation file to compare results of simulation and observed data.

1. In JavaSTICS



2. Click on "New"
3. We have 2 dates of observations
4. Select the variables for which we have observations data: see the tab Obs in the inputs_STICS_PW1 excel file

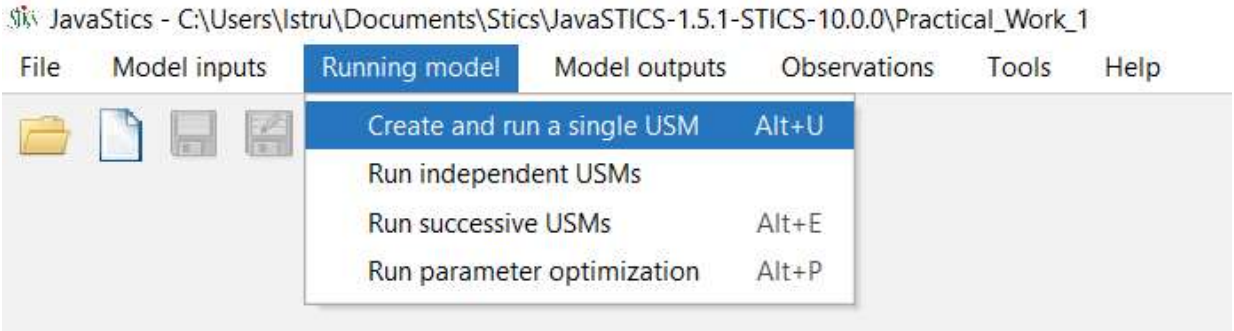


5. Click add and continue until you have selected all the variables then click on OK
6. Fill the table then click on File, Save as: Wheat_2011 (name of the USM)

ian	mo	jo	jul	HR(1)	HR(2)	HR(3)	HR(4)	AZamm(1)	AZamm(2)	AZamm(3)	AZamm(4)	AZnit(1)	AZnit(2)	AZnit(3)	AZnit(4)	resmes	ammomes	azomes	SMNmes	masecnp	mafruit
2011	3	2	61	23.39	22.36	22.07	22.40	3.1	2.85	1.31	1.01	17.07	18.64	17.77	56.33	513.25	8.27	109.82	118.09		
2011	7	21	202																	15.11	7.36

VI. Create a Unit of SiMulation: USM

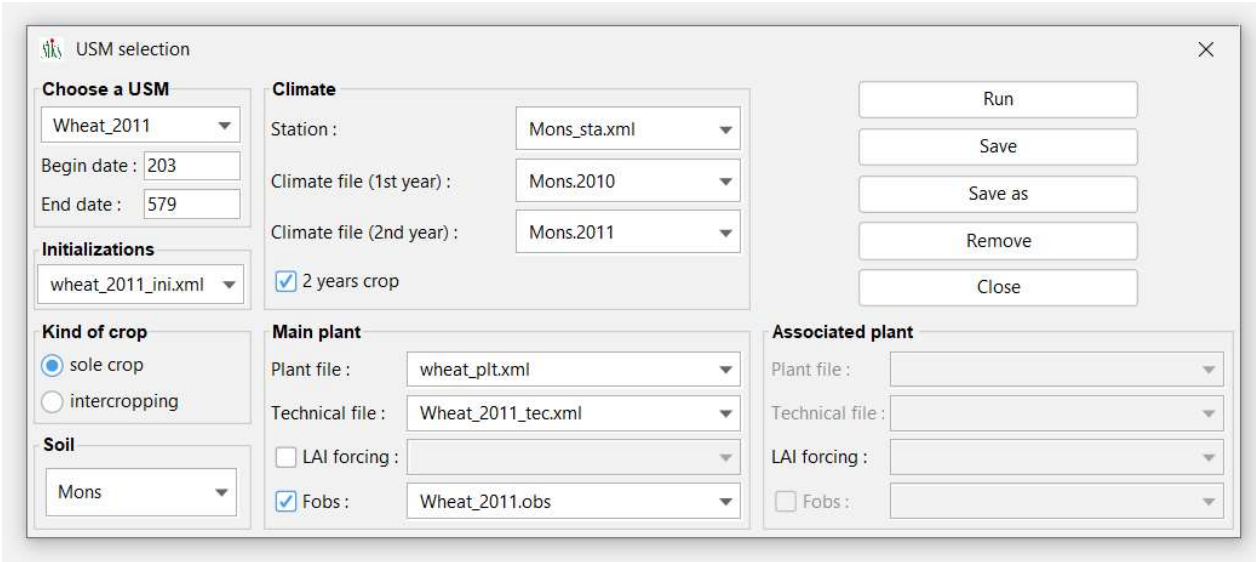
1. In JavaSTICS:



2. Create a USM with your input files from the default USM:

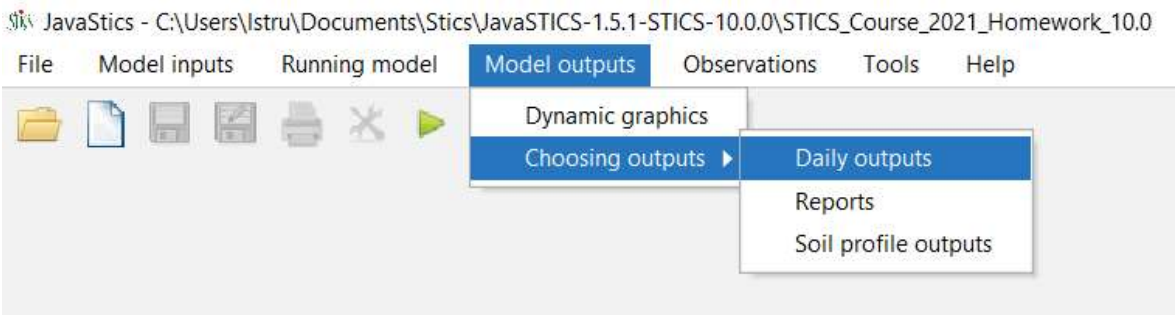
See tab "USMs" in the inputs_STICS_PW1 excel file

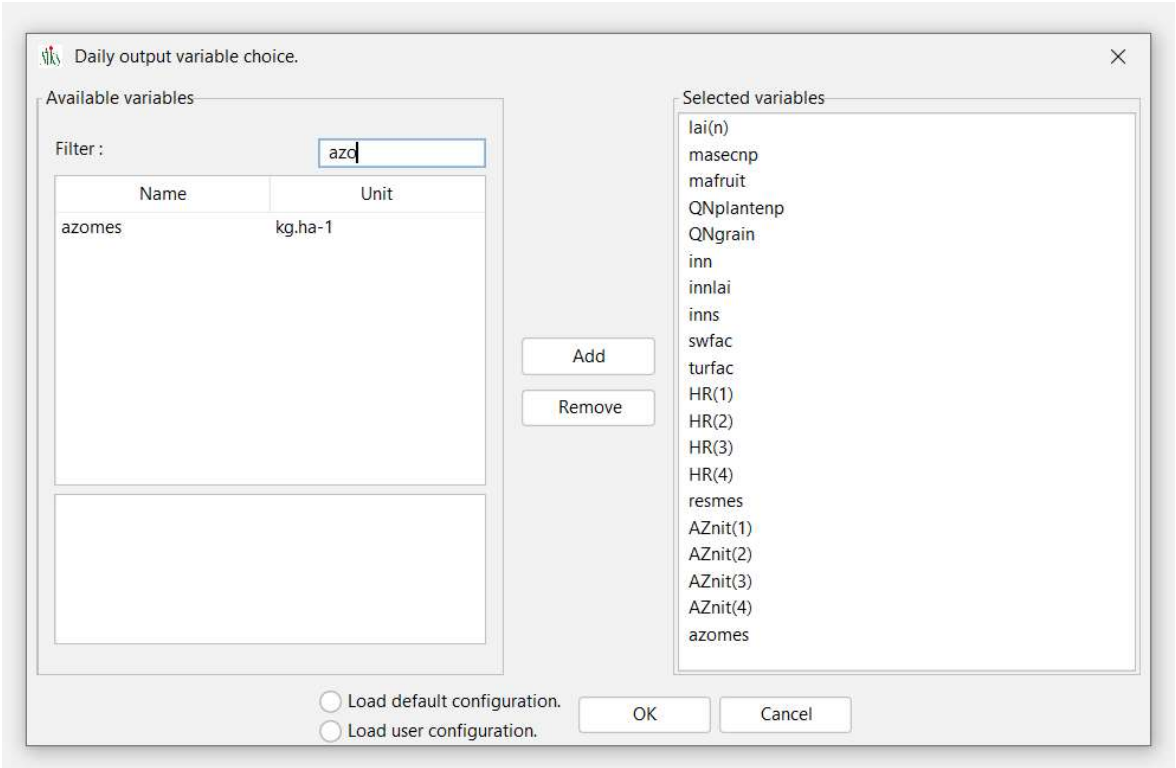
Change Begin date and End date
Select the good initialization file
We are in sole crop (nbplantes = 1)
Select the good soil
Select the good station file and associated climatic files, don't forget to check the 2 years crop box (culturean = 2)
Select the good plant file
Select the technical file
We have no file of lai observation to use the lai forcing option (flai_1 = null)
Select the observation file
Save as Wheat_2011



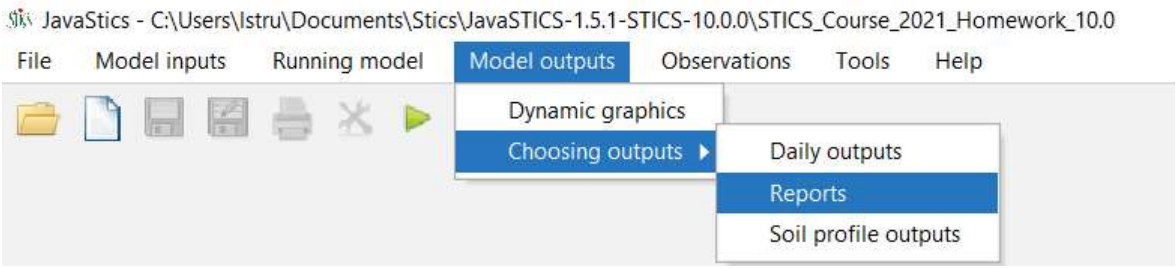
VII. Select the output data and run the USM

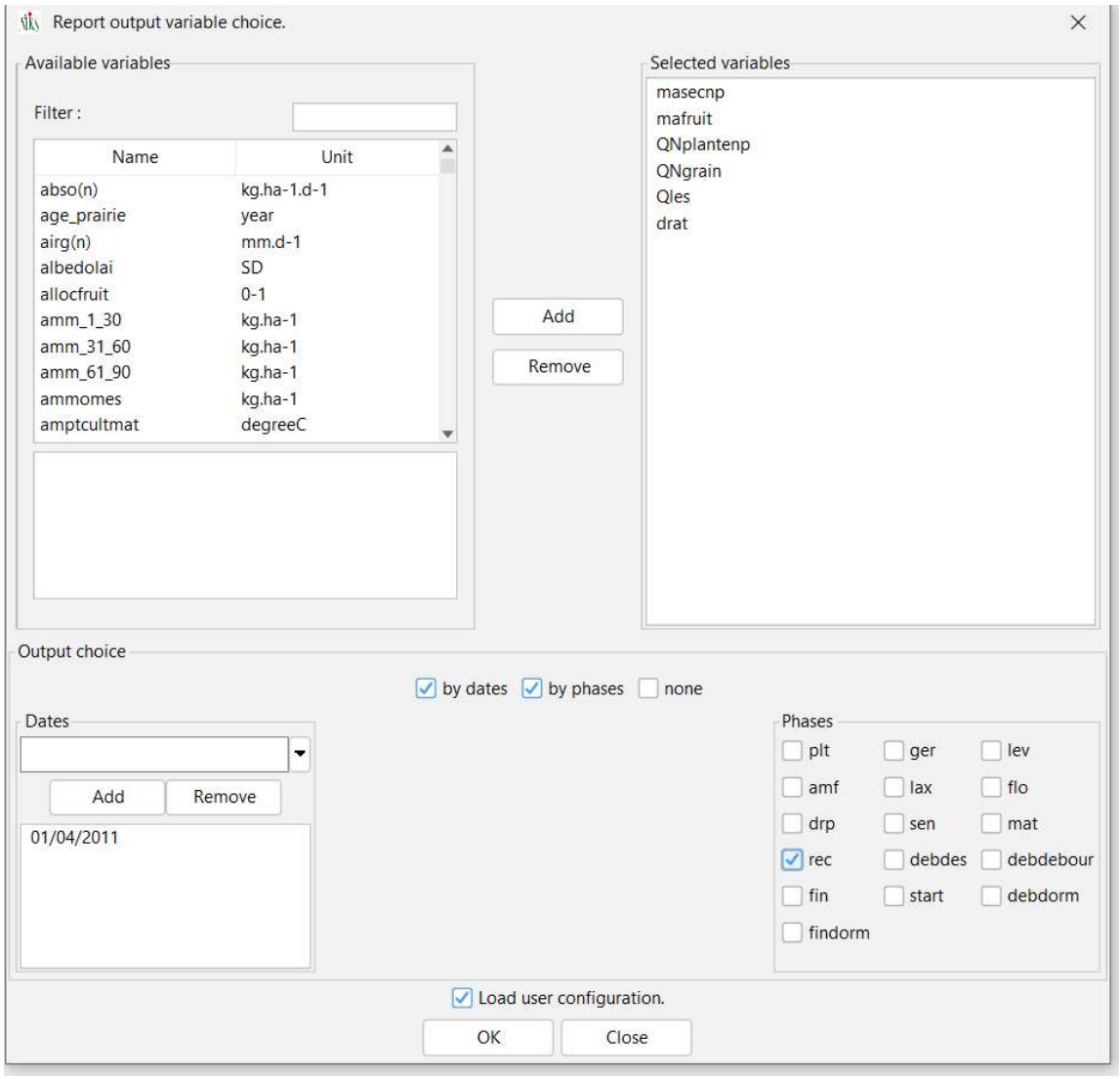
1. Selection of daily output variables in JavaSTICS



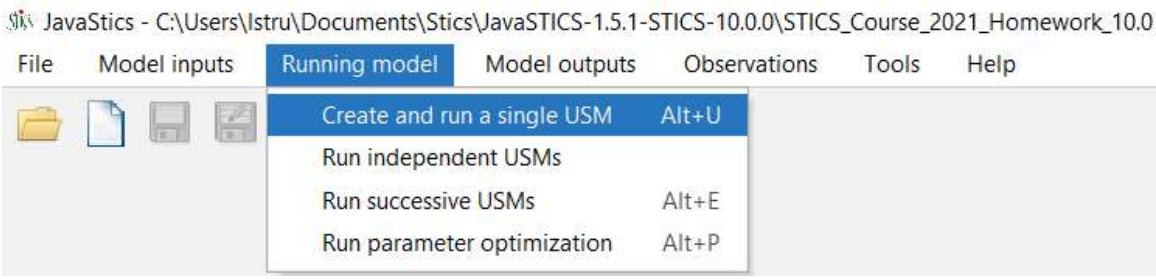


2. Selection of output variables at different dates or stages



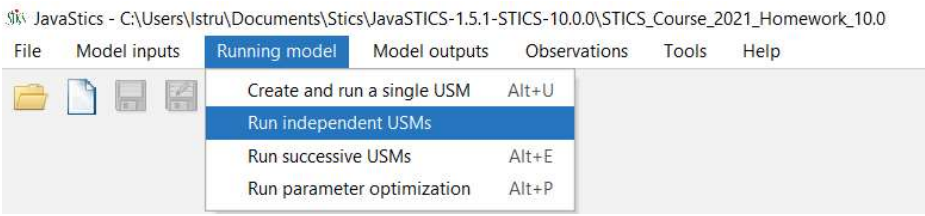


3. Run the simulation:
   - ✓ In JavaSTICS:



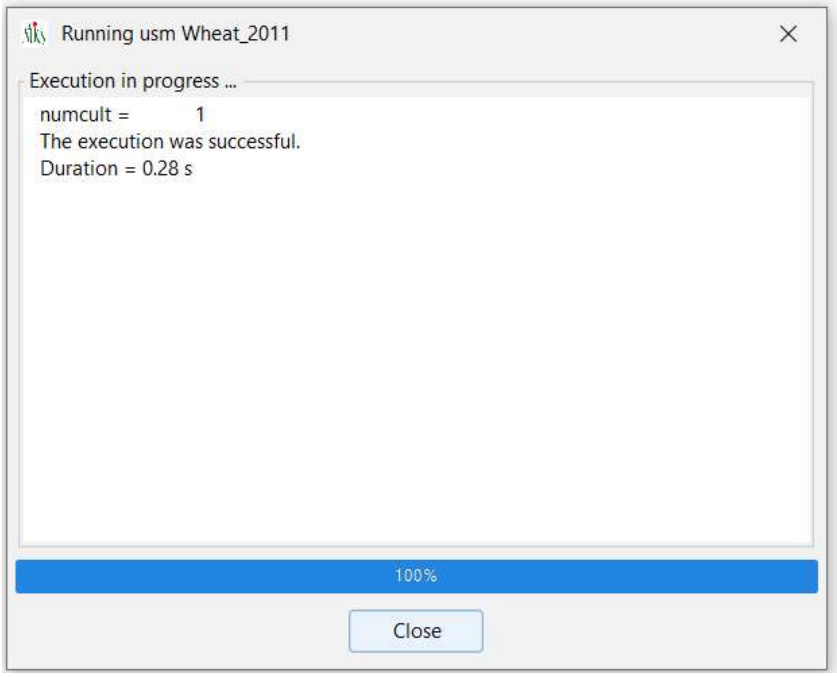
   - ✓ Select your USM
   - ✓ Run

Or:



- ✓ Select your USM
- ✓ Add
- ✓ Ok

You should have this message,



if not let me know! You did a mistake but that doesn't matter because you are here to learn!

VIII. Have a look on the model output

At the end of the simulation the model opens 2 files in the JavaSTICS interface:
modhistory.sti:
This file summarizes the parameters used for the simulation. If you have a problem during the simulation this is the only file created and it could say you why the simulation doesn't run!

mod_bWheat_2011.sti:
This file gives you the main information about the simulation results and balance of the simulation.

We should have the same results, if not, let me know!

Graphics:
Do a graphic with the JavaSTICS interface.

The files corresponding to the daily outputs and to the report are stored in the working directory Practical_Work_1:
mod_rapport.sti
mod_sWheat_2011.sti

To open the file with excel, you just have to modify the file extension .sti by .csv

This is the end! Good job!